

Paleo-Climate

Equable Climates (and taking the observations on faith)

- Much higher temperatures at continental high latitudes in Paleocene-Eocene boundary
 - Crocodilian remains (champsosaurs, lizards, tortoises etc) found in high latitudes in continental Canada, Ellesmere Island (78° N). Winter no colder than 4° C.
 - Change in vegetation types in (for example) Wyoming indicative of warm climates (Koch, Wing etc)
 - Large warming of oceanic bottom waters.
- Somewhat higher temperatures at low latitudes.
 - So apparently a much reduced pole-equator temperature gradient.

No GCM has been able to simulate such a climate. Difficulties are:

- (i) Pole—equator temperature gradient.
- (ii) Mild continental climates in winter.

Unlikely that a GCM will ever be able to provide such a simulation without some hypothesis about what is going on, because our present-day prejudices are built in.

Theory alone is unlikely to be sufficient: will not be quantitative.